

CO1 - AT1 - Part 2 Class Activities

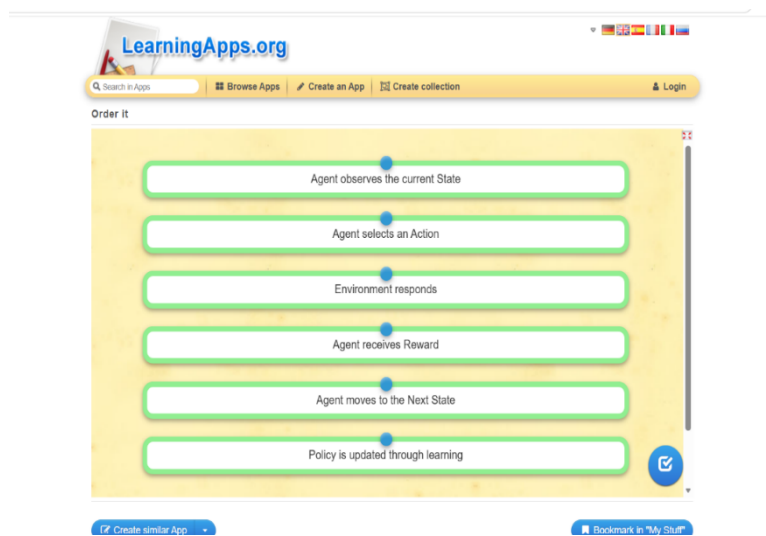
Name: P. Hithaishi

Reg no:192425196

Course Name: Reinforcement Learning

Course Code: MLA0305

Activity 1:



Activity 2:



Activity 3:

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Find the missing concepts

Reinforcement Learning is a branch of **Machine Learning** in which an **Agent** learns by interacting with an **Environment**. Instead of learning from labeled data, the agent improves its performance through trial and error by receiving **Rewards** after every action. The agent first observes the current **State**, selects an appropriate **Action**, and then moves to a new state based on the response of the environment. The strategy followed by the agent is known as a **Policy**, whose objective is to maximize the cumulative **Reward** over multiple interactions. The mathematical framework used to model this interaction is called a **Markov Decision Process**, which consists of **States**, actions, rewards, and transition probabilities. During training, the agent balances **Exploration** by trying new actions and **Exploitation** by selecting actions that have produced higher rewards in the past. Reinforcement Learning is widely used in robotics, game playing, autonomous vehicles, and intelligent decision-making systems. Most Reinforcement Learning applications are developed using **Python** programming language. Popular libraries include **TensorFlow** for deep learning, **Keras** as a high-level neural network API, **NumPy** for numerical computations, and **Gymnasium** for providing standard environments used to train and evaluate RL agents. As the number of learning **Episodes** increases, the agent gradually improves its **Policy** and makes better **Decisions** to achieve higher long-term rewards.

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Activity 4:

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Words concept mapping

O	W	D	S	P	X	E	T	H	E	Y	K	J	C	E	E	N
D	H	A	I	B	Q	X	G	F	A	Y	S	M	B	P	L	U
M	Q	L	E	Z	X	Z	D	N	L	E	A	R	N	I	N	G
K	V	E	H	R	T	J	I	X	G	U	G	K	U	S	T	T
X	U	D	U	I	V	D	P	P	O	L	I	C	Y	O	R	I
V	T	F	I	F	P	M	S	J	R	X	F	J	J	D	A	N
T	D	R	L	N	D	E	N	V	I	R	O	N	M	E	N	T
A	Z	J	C	U	M	U	L	A	T	I	V	E	I	J	S	E
F	U	Y	P	L	F	M	A	C	H	I	N	E	I	V	I	R
R	E	I	N	F	O	R	C	E	M	E	N	T	H	U	T	A
K	U	U	X	P	T	K	N	O	H	A	O	H	M	L	I	C
A	W	D	Y	J	J	K	U	J	C	B	I	N	K	K	O	T
A	U	X	K	R	G	V	O	R	K	E	A	I	A	O	N	I
S	Q	L	L	O	A	X	I	Z	G	B	E	E	B	D	N	O
H	E	A	B	Z	B	O	C	I	B	U	P	T	G	L	V	N

1. REINFORCEMENT
2. POLICY
3. CUMULATIVE
4. ENVIRONMENT
5. LEARNING
6. MACHINE
7. EPISODE
8. ALGORITHM
9. TRANSITION
10. INTERACTION

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Activity 5:

Which statement about a policy is correct?

- A** It calculates transition probabilities.
- B** It stores every reward received.
- C** It maps states to actions.
- D** It generates random environments.